VeriPromise Using in-context learning with Qwen3.5

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Task introduction

- 1. Promise Identification (PI):
  - Determine whether a segment expresses promising contents.
- 2. Supporting Evidence Assessment: assess
  - whether promises contain concrete evidence.
- 3. Clarity of the Promise–Evidence Pair
  - (CPEP): evaluate the clarity and relevance of
  - evidence in relation to the promise.
- 4. Timing for Verification (TV): indicate when
  - a promise should be revisited for verification
  - (e.g., within_two_years, two_to_five_years,
  - more_than_5_years, others).

子任務一：承諾識別

評估指標

- **Macro-F1：** 精確率與召回率的調和平均
- 衡量模型識別 ESG 承諾語句的能力

子任務二：證據支持判斷

評估指標

- **Macro-F1：** 判斷承諾是否具備充分支持證據
- 核心實務能力評估

子任務三：清晰度評估

評估指標

- **Macro-F1：** 三分類（清楚/不清楚/誤導）平均表現
- 最具挑戰性任務，辨識漂綠風險能力

子任務四：驗證時機預測

評估指標

- **Macro-F1：** 四分類時間推論能力
- 評估模型對承諾時間軸的理解

Dataset


```json
{
    "id": 10002,
    "data": "在與供應商攜手邁向永續發展的過程中，台泥致力於建立一套以合作為基礎的價值體體系。核心原則是推動供應鏈的低碳轉型，同時維護人
    "esg_type": "E",
    "promise_status": "Yes",
    "promise_string": "台泥致力於建立一套以合作為基礎的價值體體系。",
    "verification_timeline": "between_2_and_5_years",
    "evidence_status": "Yes",
    "evidence_string": "核心原則是推動供應鏈的低碳轉型，同時維護人權、保護環境和促進生物多樣性，以建立永續的合作夥伴關係。",
    "evidence_quality": "Not Clear",
    "company": "tcc",
    "ticker": 1101,
    "page_number": 32,
    "pdf_url": "https://hsxn1sjvkgtdpixe.public.blob.vercel-storage.com/tcc_1101_page_32.pdf",
    "company_source": "https://www.tccgroupholdings.com/tw/esgReport.html"
},
```

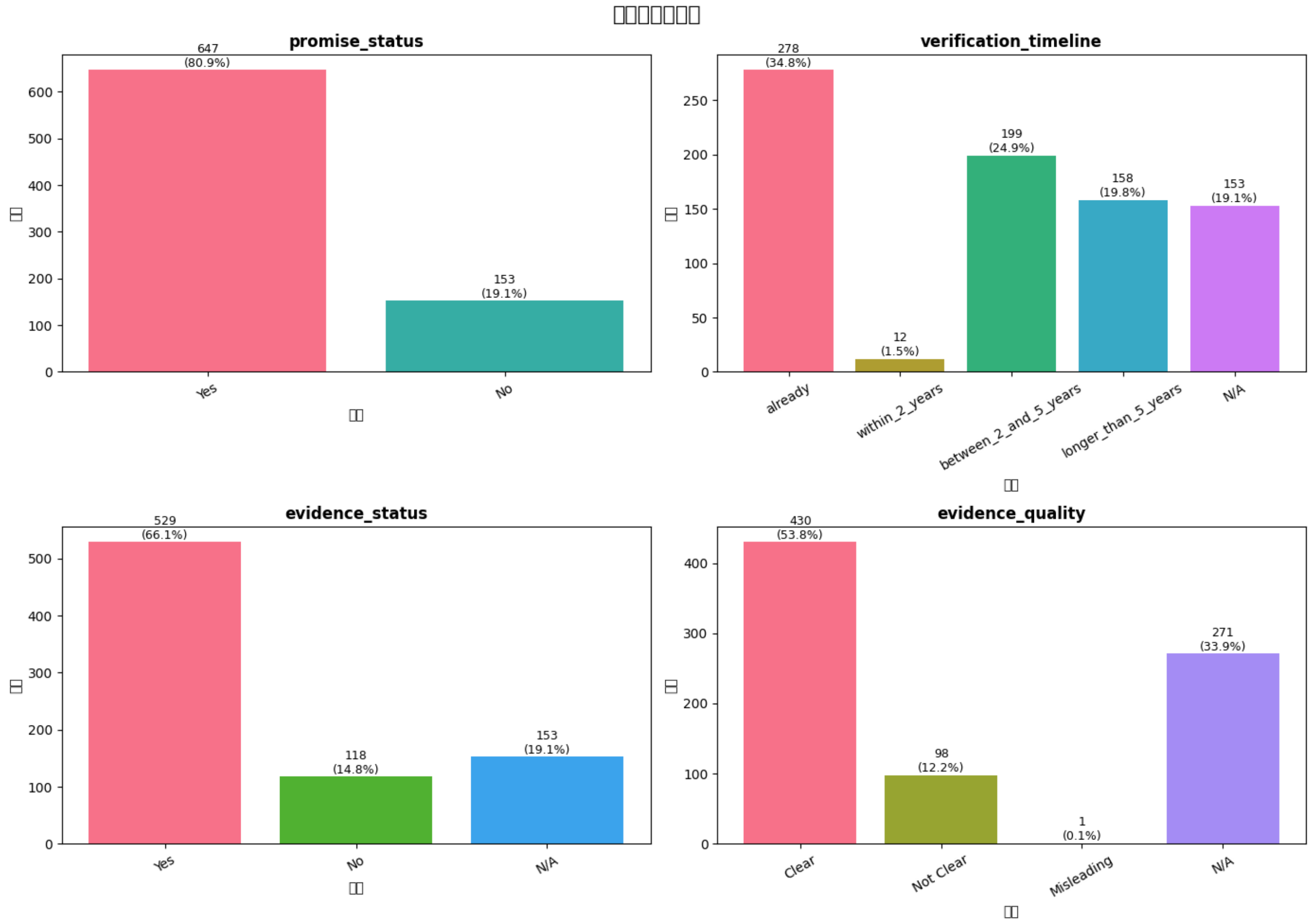
ESG type

- 環境 (Environmental, E)：關注企業對環境的影響，如溫室氣體排放（碳足跡）、能源效率、廢棄物管理、水資源管理、生物多樣性、產品綠化等。
- 社會 (Social, S)：關注企業與利害關係人的關係，如勞工權益、員工福利、工作健康與安全、多元平等（DEI）、數據隱私、產品責任、供應鏈社會責任。
- 公司治理 (Governance, G)：關注企業管理結構與倫理，如董事會組成、誠信經營、資訊透明度、風險管理、股東權利、供應鏈管理、商業道德。

Dataset


```json
{
    "id": 10001,
    "data": "聯發科技除在「工作規則」中依照勞基法明確規定「員工在產假期間公司不得終止勞動契約」外，為支持同仁與其家人度過人生不同階段，自 2024 年起提供女性員工在分娩前後計有 12 週共 84 天的產假；男性員工則可於其配偶懷孕期間陪同產檢或生（流）產日及前後 15 日間請假陪伴，兩者合計共有 10 天陪產（檢）假可運用，陪產（檢）假期間工資照常給付。",
    "esg_type": "S",
    "promise_status": "Yes",
    "promise_string": "為支持同仁與其家人度過人生不同階段，自 2024 年起提供女性員工在分娩前後計有 12 週共 84 天的產假；男性員工則可於其配偶懷孕期間陪同產檢或生（流）產日及前後 15 日間請假陪伴，兩者合計共有 10 天陪產（檢）假可運用，陪產（檢）假期間工資照常給付。",
    "verification_timeline": "already",
    "evidence_status": "No",
    "evidence_string": "",
    "evidence_quality": "N/A",
    "company": "mediatek",
    "ticker": 2454,
    "page_number": 48,
    "pdf_url": "https://hsxn1sjvkgtdpixe.public.blob.vercel-storage.com/mediatek_2454_page_48.pdf",
    "company_source": "https://corp.mediatek.tw/about/sustainability"
},
```

Data distribution



Data Preprocess

- Remove additional form
  - "esg_type ", "company ", "ticker ", "page_number, "pdf_url", "company_source“
- Data spilit
  - Train:Test = 8:2

Load model

- Qwen/Qwen3.5-4B

- Type: Causal Language Model with Vision Encoder
- Training Stage: Pre-training & Post-training
- Language Model
  - Number of Parameters: 4B
  - Hidden Dimension: 2560
  - Token Embedding: 248320 (Padded)
  - Number of Layers: 32

Qwen3.5

- https://huggingface.co/Qwen/Qwen3.5-4B

Prompting

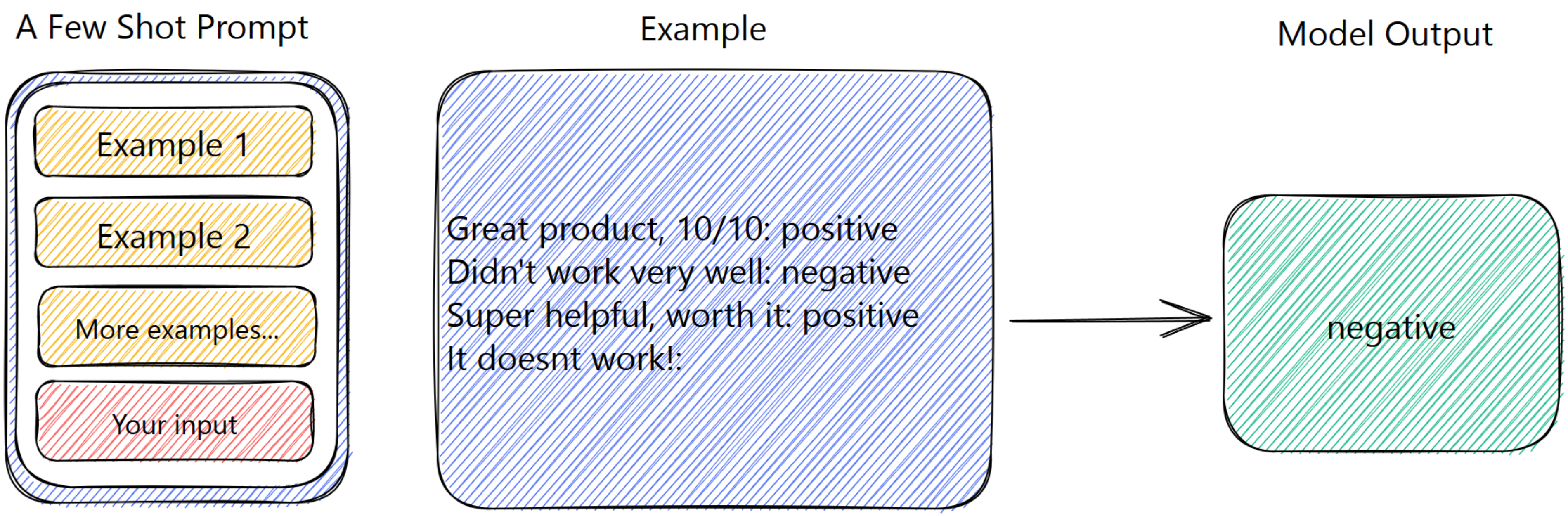


Image source

Prompting

```
Prompt 1: Field definitions:
1. promise_status: Whether the text contains an ESG promise/commitment. Values: Yes | No
2. evidence_status: Evaluate whether commitment statements are semantically clear and unambiguous, and
Values: Yes | No | N/A
3. evidence_quality: Quality of the evidence provided.
   Values: Clear | Not Clear | Misleading | N/A
4. verification_timeline: When the promise is expected to be fulfilled.
   Values: already | within_2_years | between_2_and_5_years | longer_than_5_years | N/A

--- Examples ---
Text: 遠傳電信視連續兩年有交易之供應商為有效供應商，2024 年度有效供應商為 1,070 家，其中客戶裝置與資通訊類採購
promise_status: No
evidence_status: N/A
evidence_quality: N/A
verification_timeline: N/A

Text: 為減少各項能源耗用，合庫金控導入「ISO 50001 能源管理系統」，透過各項自發性之行動計畫與改善方案，致力節能、
promise_status: Yes
evidence_status: Yes
evidence_quality: Clear
verification_timeline: already
```

Predict

```
for idx, item in enumerate(test_set):
    examples = rng.sample(train_set, FEW_SHOT_K)
    prompt   = build_few_shot_prompt(examples, item)
    output   = generate(prompt)
    preds    = parse_output(output)
```

```
Prompt 1: {'promise_status': 'Yes', 'verification_timeline':
'already', 'evidence_status': 'Yes', 'evidence_quality':
'Clear'}
```

Scoring

最終分數 = 加權平均 Macro F1：

- promise_status × 0.20
- evidence_status × 0.30
- evidence_quality × 0.35
- verification_timeline × 0.15

```
===== Evaluation Results =====
  promise_status              Macro F1 = 0.6893  (weight=0.2)
  verification_timeline       Macro F1 = 0.4243  (weight=0.15)
  evidence_status             Macro F1 = 0.5959  (weight=0.3)
  evidence_quality            Macro F1 = 0.3370  (weight=0.35)

  Weighted Macro F1 Score = 0.4982
```

Sample Code

- https://github.com/Bruceliangcf/veripromise_ICL